Answer Key MCQs

Decision Trees (Entropy and Information Gain):

- Q1: A
- Q2: A, B, C
- Q3: A, C, D

Random Forests:

- Q4: B
- Q5: A, B, C
- Q6: A, C

Overfitting/Underfitting + Bias/Variance:

- Q7: A, B, C
- Q8: A, B, C, D
- Q9: B, C, D

Bagging (Bootstrap Aggregation):

- Q10: A, B, C
- Q11: A, C, D
- Q12: A, C

Boosting:

- Q13: A, C, D
- Q14: A, B, D
- Q15: A, C, D

Isolation Forests:

- Q16: A, B, D
- Q17: A, B, C, D
- Q18: A, B, D

Answer Key Numericals

Q19.

A.

Counts: $n_1 = 5, n_0 = 3,\ p_1 = 5/8, p_0 = 3/8$

$H(Y) = -\sum p_k log_2 p_k \approx 0.9544$

B. Left = $\{0, 0, 0\}$ (entropy 0), Right = $\{1, 1, 1, 1, 1\}$ (entropy 0).
Weighted child entropy = 0. Information Gain = 0.9544 - 0 = 0.95444.

C. Left = $\{0, 0, 0, 1, 1\}$ (entropy 0.9709), Right = $\{1, 1, 1, 1, 1\}$ (entropy 0).
Weighted child entropy = 0.6068. Information Gain = 0.9544 - 0.6068 = 0.3476.

D. Better split: $x \leq 4.5$

Q20.

A. $1 - (0.6^2 + 0.4^2) = 0.48$

B. Left: $1 - (0.8^2 + 0.2^2) = 0.32$, Right: $1 - (0.4^2 + 0.6) = 0.48$

C. $0.5 \cdot 0.32 + 0.5 \cdot 0.48 = 0.40$

D. $0.48 - 0.4 = 0.08$

SVM

1. Identify the Support Vectors

To find the support vectors, we must find the closest pair of points from opposite classes.

The two classes are {P_1(3, 4), P_3(2, 5)} and {P_2(5, 2)}. We need to calculate the Euclidean distance for the two possible pairs of opposite-class points: (P_1, P_2) and (P_3, P_2).

- **Distance between P_1(3, 4) and P_2(5, 2):** d = $\sqrt{((5-3)^2 + (2-4)^2)} = \sqrt{(2^2 + (-2)^2)} = \sqrt{(4+4)} = \sqrt{8} \approx 2.828$
- **Distance between P_3(2, 5) and P_2(5, 2):** d = $\sqrt{((5-2)^2 + (2-5)^2)} = \sqrt{(3^2 + (-3)^2)} = \sqrt{(9+9)} = \sqrt{18} \approx 4.243$

Since the distance between P_1 and P_2 is smaller, these are the two points that define the margin.

The support vectors are P_1(3, 4) and P_2(5, 2).

2. Calculate the Equation of the Decision Boundary

The decision boundary is defined by its weight vector **w** and bias **b**.

- **Weight Vector (w):** The weight vector **w** is parallel to the vector connecting the two support vectors.
    - Vector between SVs: $P_1 - P_2$ = (3-5, 4-2) = (-2, 2).
    - We can use a proportional vector for simplicity: **w = (-1, 1)**.
- **Bias (b):** The hyperplane lies exactly halfway between the two support vectors. We find the midpoint of the support vectors and use it to solve for *b*.
    - **Midpoint M:** ((3+5)/2, (4+2)/2) = (4, 3).
    - This midpoint must lie on the decision boundary, so it satisfies the equation $w^Tx - b = 0$.
    - Substitute M(4, 3) into the equation: (-1)(4) + (1)(3) - b = 0 -4 + 3 - b = 0 -1 - b = 0 **b = -1**

The equation of the decision boundary is $-x_1 + x_2 + 1 = 0$.

3. Calculate the Width of the Margin

The total width of the margin is given by the formula 2 / ||w||.

- First, calculate the magnitude (L2-norm) of our weight vector **w = (-1, 1)**:
    - $||w|| = \sqrt{((-1)^2 + (1)^2)} = \sqrt{(1 + 1)} = \sqrt{2}$.
- Calculate the margin width:
    - **Margin Width** = $2 / \sqrt{2} = \sqrt{2} \approx 1.414$.

Linear Regression

A. Slope (b) = 0.425

B. Intercept (a) = 0.785

C. Equation of fitted line:

$$\hat{y} = 0.785 + 0.425x$$

D. Predicted value at x = 6:

$$\hat{y}(6) = 3.335$$

E. Sum of Squared Errors (SSE):

$$SSE = 2.7907$$

SVM with Polynomial Kernel

A. Kernel matrix

K=

4	4	0	0
4	4	0	0
0	0	4	4
0	0	4	4

B. Lagrange multipliers

$\alpha1=\alpha2=\alpha3=\alpha4=1/8$.

C. Bias term

b=0

D. Decision function

$f(x)=\sum_{i=1}^{4}\alpha_i y_i K(x_i,x)+b = 1/8\sum_{i=1}^{4} y_i K(x_i,x)$

E. Prediction for x=(0.5,0.5)

Compute $K(x1,x)=1$, $K(x2,x)=1$, $K(x3,x)=0$, $K(x4,x)=0$.

$f(x)=\frac{1}{8}(1\cdot 1+1\cdot 1+(-1)\cdot 0+(-1)\cdot 0)=0.25$.

Predicted class=+1.

Decision Trees (Entropy & Information Gain)

Solution 1:

1. **Information gain for Feature A (Study Hours):**

   Parent entropy:

   - E(parent) = $-(28/40)\log_2(28/40) - (12/40)\log_2(12/40)$
   - E(parent) = -(0.7)(-0.515) - (0.3)(-1.737)
   - **E(parent) = 0.361 + 0.521 = 0.882**
2. **Child entropies:**

   - Low: 5P, 8F (13 total)
     - E(Low) = $-(5/13)\log_2(5/13) - (8/13)\log_2(8/13)$
     - **E(Low) = 0.961**
   - Medium: 10P, 3F (13 total)
     - E(Med) = $-(10/13)\log_2(10/13) - (3/13)\log_2(3/13)$
     - **E(Med) = 0.779**
   - High: 13P, 1F (14 total)
     - E(High) = $-(13/14)\log_2(13/14) - (1/14)\log_2(1/14)$
     - **E(High) = 0.371**
3. **Weighted average:**

   - Weighted = (13/40)(0.961) + (13/40)(0.779) + (14/40)(0.371)
   - Weighted = 0.312 + 0.253 + 0.130 = 0.695

4. **Gain(A) = 0.882 - 0.695 = 0.187**

5. **Information gain for Feature B (Attendance):**

   Child entropies:

   - Poor: 3P, 7F (10 total)
     - E(Poor) = $-(3/10)\log_2(3/10) - (7/10)\log_2(7/10)$
     - **E(Poor) = 0.881**
   - Good: 12P, 4F (16 total)
     - E(Good) = $-(12/16)\log_2(12/16) - (4/16)\log_2(4/16)$
     - **E(Good) = 0.811**
   - Excellent: 13P, 1F (14 total)
     - **E(Excellent) = 0.371** (same as High study hours)
6. **Weighted average:**

   - Weighted = (10/40)(0.881) + (16/40)(0.811) + (14/40)(0.371)
   - Weighted = 0.220 + 0.324 + 0.130 = 0.674
7. **Gain(B) = 0.882 - 0.674 = 0.208**

8. **Best feature selection:**

   - **Feature B (Attendance) should be selected** as it has higher information gain (0.208 vs 0.187)
9. **Highest remaining impurity:**

   - After splitting on Attendance, the node with highest entropy is **"Poor Attendance" with E = 0.881**

Random Forests

Problem 1: Bootstrap Sampling Analysis

Solutions:

1. **Instances never selected in Tree 1:**

   - Tree 1 sample: {A, A, C, D, E, E, E, G, I, J}
   - Original: {A, B, C, D, E, F, G, H, I, J}
   - **Never selected: B, F, H**
2. **Final prediction:**

   - Tree 1: Positive
   - Tree 2: Negative

    - Tree 3: Positive
    - **Final prediction: Positive** (2 votes vs 1 vote)
3. **Confidence level:**

    - **Confidence = 2/3 = 66.67%**

Problem 2: Forest Prediction Aggregation

Solutions:

1. **Final classification:**

    - Disease votes: 5 (Trees 1, 3, 4, 6, 7)
    - Healthy votes: 2 (Trees 2, 5)
    - **Final classification: Disease**
2. **Prediction confidence:**

    - **Confidence = 5/7 = 71.43%**
3. **Treatment recommendation:**

    - **Yes, proceed with treatment recommendation**
    - The confidence (71.43%) exceeds the required threshold of 70%
4. **Trees needed to reverse:**

    - Currently: 5 Disease, 2 Healthy
    - To reverse (make Healthy win): need at least 4 Healthy votes
    - **Need to change 2 trees** from Disease to Healthy
    - This would make it 3 Disease, 4 Healthy

Overfitting/Underfitting + Bias/Variance

Problem 1: Model Performance Comparison

Solution 1:

1. **Training-validation gaps:**

    - **Model A:** 73% - 71% = 2%
    - **Model B:** 89% - 86% = 3%
    - **Model C:** 99% - 68% = 31%
2. **Overfitting evidence:**

- **Model C shows clear signs of overfitting**
- Evidence:
    - Extremely high training accuracy (99%) but poor validation accuracy (68%)
    - Large training-validation gap of 31 percentage points
    - The model has essentially memorized the training data but fails to generalize

3. **Deployment recommendation:**

    - **Deploy Model B**
    - Reasoning:
        - Highest validation accuracy (86%)
        - Small training-validation gap (3%) indicates good generalization
        - Balanced performance between fitting and generalizing

4. **Highest variance:**

    - **Model C has the highest variance**
    - The 31% gap between training and validation indicates the model is highly sensitive to the specific training data and captures noise rather than true patterns

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Problem 2: Performance Degradation Analysis

Solutions:

1. **Performance degradation:**

    - Validation at depth 10: 83%
    - Validation at depth 15: 77%
    - **Degradation = 83% - 77% = 6 percentage points**

2. **When overfitting starts:**

    - **Overfitting becomes problematic after depth 10**
    - Evidence:
        - Depth 5→10: Validation improves (+3%)
        - Depth 10→15: Validation decreases (-6%)
        - At depth 15, training continues to improve (+5%) while validation drops

3. **Choice between depth 10 and depth 5:**

    - **Choose depth 10**
    - Reasoning:
        - Validation accuracy is higher (83% vs 80%)
        - Training-validation gap is still reasonable (11%)
        - The model at depth 5 is likely underfitting (both accuracies relatively low)

- Depth 10 provides the best balance before overfitting begins

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Isolation Forests

Problem 1: Path Length Calculation

Solutions:

1. **E(h(X)) - Average path length:**
    - Tree 1: 4 edges
    - Tree 2: 3 edges
    - Tree 3: 5 edges
    - **E(h(X)) = (4 + 3 + 5) / 3 = 12 / 3 = 4**
2. **Anomaly score:**
    - Formula: s(X, n) = 2^(-E(h(X))/c(n))
    - s(X, 100) = 2^(-4/10.29)
    - s(X, 100) = 2^(-0.389)
    - **s(X, 100) ≈ 0.763**
3. **Is X an anomaly?**
    - Anomaly score: 0.763
    - Threshold: s > 0.6
    - **Yes, instance X is likely an anomaly** because 0.763 > 0.6
    - The relatively short average path length (4) compared to expected path for normal instances indicates X was isolated quickly, suggesting it's an outlier

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Problem 2: Isolation Forest Evaluation

given:

- 4 trees, n = 200, c(n) = 10.66
- Formula: s(x,n) = 2^(-E(h(x))/c(n))

1. E(h(X)) for each instance:

P: (8+7+9+8)/4 = **8.0**

Q: (3+2+4+3)/4 = **3.0**

R: (6+5+7+6)/4 = **6.0**

2. Anomaly scores:

P: s = 2^(-8.0/10.66) = 2^(-0.751) = **0.594**

Q: s = 2^(-3.0/10.66) = 2^(-0.281) = **0.823**

R: s = 2^(-6.0/10.66) = 2^(-0.563) = **0.676**

3. Ranking (most to least anomalous):

1. **Q (0.823)** - Most anomalous
2. **R (0.676)** - Moderately anomalous
3. **P (0.594)** - Likely normal

4. Top 30% flagging:

Top 30% of 3 instances = 1 instance

Flag: Instance Q only

5. Path length vs anomaly score relationship:

Inverse relationship: Shorter path → Higher score → More anomalous

Explanation:

- Anomalies have unusual values → isolated quickly (few splits needed)
- Normal instances blend with others → isolated slowly (many splits needed)
- Q: path=3.0 → isolated fast → anomaly
- P: path=8.0 → isolated slow → normal